SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Coding & Solution

Date	29 June 2026
Team ID	N/A
Project Name	OptiCrop Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/neelimavana/OptiCrop
Programming Language(s)	Python, HTML5, CSSS
Framework(s) Used	Flask
Machine Learning Library	Scikit-learn (Random Forest Classifier)
Data Analysis Libraries	NumPy, Pandas
Visualization Libraries	Matplotlib, Seaborn
Key Features Implemented	• AI-based Crop Recommendation System • Crop Suitability Assessment • Soil & Environmental Parameter Analysis • User-friendly Flask Web Interface • Machine Learning-based Prediction Engine • Agricultural Production Optimization

Field	Details
Pending / Incomplete Features	• User Authentication & Login System • Database Integration (MySQL/SQLite) • Cloud Deployment (Render/AWS/Heroku) • Fertilizer Recommendation Module • Weather API Integration
Setup / Run Instructions	• Install Python 3.x. • Install required packages using pip install -r requirements.txt. • Open the project folder in Visual Studio Code. • Navigate to the project directory. • Run the command:python app.py • Open the local URL displayed in the terminal (e.g., http://127.0.0.1:5000).

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions/modules	**Yes**
2	Meaningful variable and function names are used	**Yes**
3	Code includes comments/documentation where necessary	**Yes**
4	Input validation and error handling are implemented	**Yes**
5	Machine Learning model is saved and reused (.pkl)	**Yes**
6	The Flask application runs without critical errors	**Yes**
7	Source code is committed to a GitHub repository	**Yes**
8	Project follows a modular folder structure	**Yes**
9	HTML templates and static files are organized separately	**Yes**
10	Project documentation (README, architecture, diagrams) is included	**Yes**